THE EVOLUTIONARY ARC OF MACHINE LEARNING SYSTEMS

From disembodied cloud perception and edge micro-sensing to closed-loop actuation under physical laws

ERA 1 · CLOUD

Disembodied ML

2012 – 2020 · Cloud GPU

Systems Paradigm:

Stateless digital perception

- AlexNet & ResNet backbones
- Transformer attention (NLP)
- Data-parallel GPU clusters
- MLPerf benchmark suites

Execution Boundary:

Behind glass in cloud containers

- Errors caught with try/catch
- HTTP 500 retry semantics
- World state W_t unchanged

Physical Limitation:

Purely digital software;
zero interaction with kinetic
mass, energy, or matter.

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ERA 2 · TINYML

Edge Micro-Sensing

2018 – 2023 · Microcontroller

Systems Paradigm:

Ultra-low-power edge inference

- INT8 quantization & pruning
- Kilobytes of SRAM memory
- Milliwatt power budgets
- TinyMLPerf benchmark suite

Deployment Mode:

On-device acoustic & IMU sensing

- Keyword spotting (KWS)
- Anomaly vibration monitors
- Micro-vision presence detect

Physical Limitation:

Open-loop sensing only;
models emit digital flags
without closing control loops.

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ERA 3 · FOUNDATION

Semantic Reasoning

2023 – 2026 · VLM / VLA

Systems Paradigm:

Open-world semantic scene parsing

- Multimodal vision-language
- Spatial affordance grounding
- Trajectory diffusion policies
- Billions of parameters

Operating Cadence:

Slow deliberation on host NPUs

- Slow cadences: 0.5–2 Hz
- Tail latency spikes (P99)
- High power draw (15–50W)

Physical Limitation:

No physical safety reflex;
lacks awareness of inertia,
bus stalls, or clock jitter.

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ERA 4 · PHYSICAL AI

Closed-Loop Actuation

2026+ · Dual-Brain Systems

Systems Paradigm:

Delegated physical authority

- Proposal–Permission Split
- Multi-rate asynchronous IPC
- 1 kHz Control Barrier Functions
- Dynamic stopping physics (d_{stop})

Dual-Brain Silicon:

Untrusted MPU + Trusted MCU

- Expiring Intent Leases (L_{intent})
- Survives Linux kernel crashes
- Matter & Momentum Mutate (W_{t+1})

Core Breakthrough:

Open-world semantic intelligence
caged within deterministic,
provable physical safety laws.